

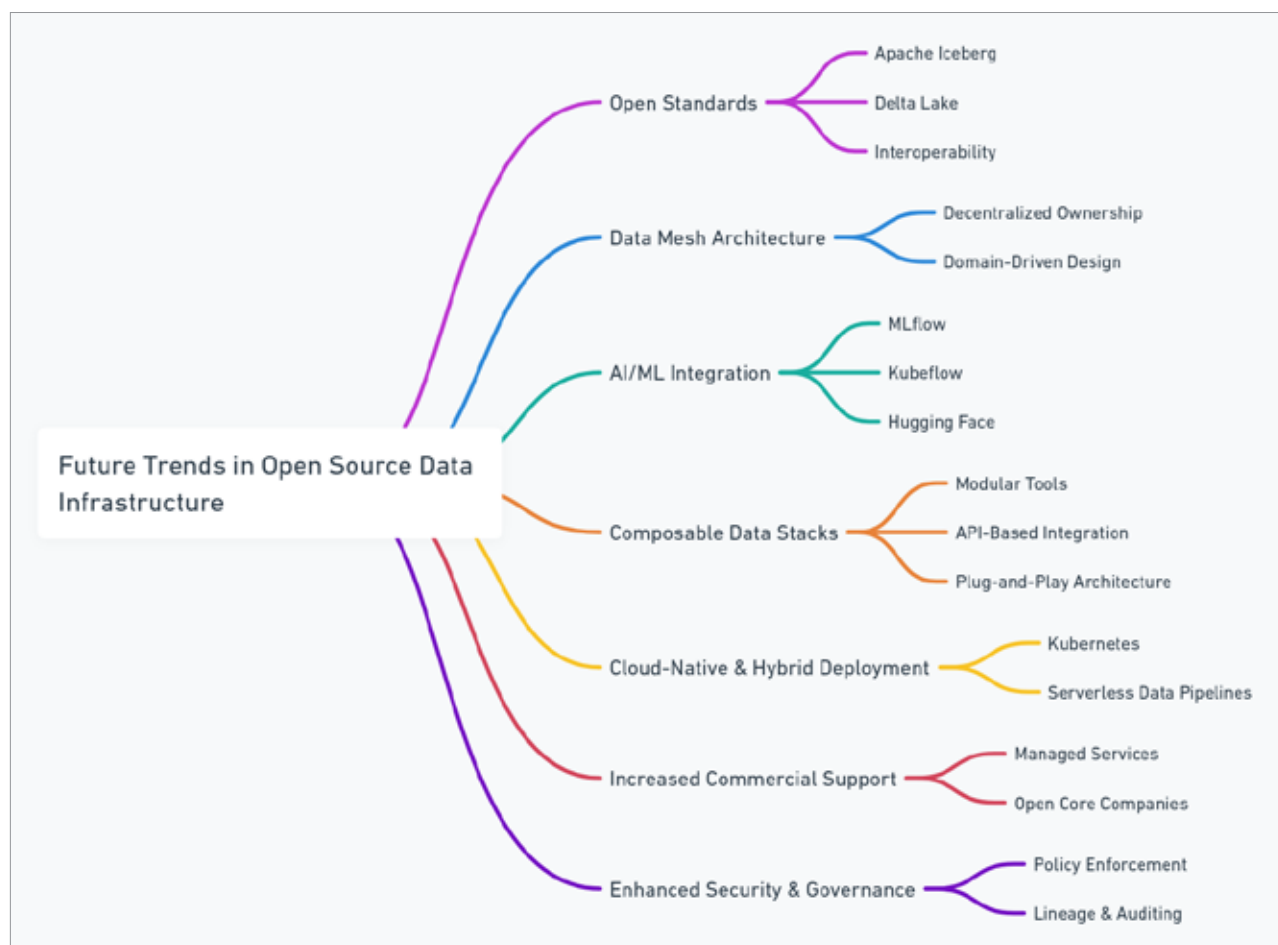


Figure 1: Open source data infrastructure: What the future holds

are typically free to use, which significantly lowers the entry barrier for startups and small enterprises. This accessibility allows organisations to experiment freely and scale without being burdened by licensing fees. While operational costs (infrastructure, talent, support) still exist, the total cost of ownership is often much lower than using proprietary solutions.

Rapid innovation and community-driven development:

Open source thrives on global collaboration. Thousands of developers contribute to the continuous improvement of data tools. Projects like Apache Kafka, initially developed at LinkedIn, and Apache Spark, from UC Berkeley, evolved rapidly thanks to widespread community support. Features are developed faster, bugs are fixed quickly, and innovation is not limited to a single vendor's roadmap.

Flexibility and customisation: Modern data challenges are not one-size-fits-all. Open source gives engineering teams full control over their stack. You can:

- Modify code for performance optimisation.
- Integrate with proprietary internal systems.

- Extend tools to support new data formats or protocols. This level of customisation is almost impossible with closed-source vendors.

Avoiding vendor lock-in: Proprietary data platforms often come with the risk of vendor lock-in, making it difficult or expensive to migrate data or switch providers. Open source provides vendor neutrality, giving teams the freedom to build hybrid or multi-cloud architectures and avoid dependency on a single company's platform or pricing model.

Security and transparency: In open source, anyone can inspect the source code. This visibility promotes transparency and trust. Vulnerabilities can be identified and patched by the global community long before a proprietary vendor may issue a fix. Also, organisations can apply their own security policies or integrations, ensuring better compliance and auditing capabilities.

Core open source technologies in data infrastructure

To manage the full data lifecycle, organisations use a wide